

Karnaugh Maps: Worked Simplification Examples

SOP, POS, redundant grouping, multiple minimal forms and don't-cares

K-map convention used throughout

Adjacent column and row labels use Gray-code order: 00, 01, 11, 10. Adjacent cells differ in only one variable. The first and last rows/columns are also adjacent.

For SOP: group 1s. For POS: group 0s.

Each drawn rounded outline identifies one K-map group. When a group wraps around an edge, it appears as two partial outlines on opposite edges of the map.

1. $F(x,y,z) = \text{Sum of minterms } (3,4,6,7)$

The 1s are placed at m3, m4, m6 and m7. We first make the two groups needed to cover all the 1s.

$x \setminus yz$	00	01	11	10
0	0 m0	0 m1	1 m3	0 m2
1	1 m4	0 m5	1 m7	1 m6

Group m3,m7: $y=1$ and $z=1$ remain unchanged $\rightarrow yz$

Group m4,m6: $x=1$ and $z=0$ remain unchanged $\rightarrow xz'$

Therefore: $F = yz + xz'$

A legal but redundant extra group

We may also group m6 and m7. In these two cells, $x=1$ and $y=1$ remain unchanged, so this extra group gives xy . But m6 is already covered by xz' and m7 is already covered by yz . The extra group therefore covers no new 1-cell and does not simplify the result further.

$x \setminus yz$	00	01	11	10
0	0 m0	0 m1	1 m3	0 m2
1	1 m4	0 m5	1 m7	1 m6

With redundant group: $F = yz + xz' + xy$ $\rightarrow$ still simplifies to $F = yz + xz'$

Why m2 and m3 cannot be grouped in Problem 1

The stated function contains m3, m4, m6 and m7 only. Hence m2 is a 0-cell. A group used for SOP simplification may contain 1s (and, when explicitly given, don't-care cells), but it cannot include an ordinary 0.

x \ yz	00	01	11	10
0	0 m0	0 m1	1 m3	0 m2
1	1 m4	0 m5	1 m7	1 m6

The dashed outline above shows the proposed m2,m3 pair. Since m2=0, this is not a valid group for the stated function. If m2 were specified as a don't-care, then m2 and m3 could be grouped and would produce $x'y$.

Note For demonstrating redundancy without introducing don't-cares, use the legal extra group m6,m7 from the previous page.

2. $F(x,y,z) = \text{Sum of minterms } (1,2,3,5,6,7)$

The largest convenient groups are two groups of four. Overlap is allowed, and m3 and m7 belong to both groups.

$x \backslash yz$	00	01	11	10
0	0 m0	1 m1	1 m3	1 m2
1	0 m4	1 m5	1 m7	1 m6

Group m1,m3,m5,m7: $z=1$ remains unchanged, so the group gives z .

Group m2,m3,m6,m7: $y=1$ remains unchanged, so the group gives y .

Therefore: $F = y + z$

A useful check: the only 0-cells are m0 and m4. In both of them $y=0$ and $z=0$, which is exactly when $y+z$ is 0.

3. Four-variable example

$F(A,B,C,D) = \text{Sum of minterms } (0,2,3,5,7,8,9,10,11,13,15)$

AB \ CD	00	01	11	10
00	1 m0	0 m1	1 m3	1 m2
01	0 m4	1 m5	1 m7	0 m6
11	0 m12	1 m13	1 m15	0 m14
10	1 m8	1 m9	1 m11	1 m10

Corner group m0,m2,m8,m10 (wrap-around): $B=0$ and $D=0$ remain unchanged $\rightarrow B'D'$.

Group m5,m7,m13,m15: $B=1$ and $D=1$ remain unchanged $\rightarrow BD$.

These two terms, $B'D'$ and BD , occur in all four minimal answers.

Two independent choices remain

For the remaining 1-cells, one may choose $A B'$ or AD for one region, and $B'C$ or CD for the other region. Combining the two independent choices gives four equally simple Boolean expressions.

Problem 3: the four possible minimal expressions

Choice 1

AB \ CD	00	01	11	10
00	1 m0	0 m1	1 m3	1 m2
01	0 m4	1 m5	1 m7	0 m6
11	0 m12	1 m13	1 m15	0 m14
10	1 m8	1 m9	1 m11	1 m10

$$F = B'D' + BD + AB' + B'C$$

Choice 2

AB \ CD	00	01	11	10
00	1 m0	0 m1	1 m3	1 m2
01	0 m4	1 m5	1 m7	0 m6
11	0 m12	1 m13	1 m15	0 m14
10	1 m8	1 m9	1 m11	1 m10

$$F = B'D' + BD + AB' + CD$$

Choice 3

AB \ CD	00	01	11	10
00	1 m0	0 m1	1 m3	1 m2
01	0 m4	1 m5	1 m7	0 m6
11	0 m12	1 m13	1 m15	0 m14
10	1 m8	1 m9	1 m11	1 m10

$$F = B'D' + BD + AD + B'C$$

Choice 4

AB \ CD	00	01	11	10
00	1 m0	0 m1	1 m3	1 m2
01	0 m4	1 m5	1 m7	0 m6
11	0 m12	1 m13	1 m15	0 m14
10	1 m8	1 m9	1 m11	1 m10

$$F = B'D' + BD + AD + CD$$

4. Product-of-Sums (POS) example

Example: $F(A,B,C) = \text{Product of maxterms } (0,2,5,7)$. For POS simplification, we group the 0-cells rather than the 1-cells.

A \ BC	00	01	11	10
0	0 m0	1 m1	1 m3	0 m2
1	1 m4	0 m5	0 m7	1 m6

Zero-group m0,m2 (wrap-around): $A=0$ and $C=0$ remain unchanged. For a POS sum term, a constant 0 appears uncomplemented. Hence this group gives $(A + C)$.

Zero-group m5,m7: $A=1$ and $C=1$ remain unchanged. For a POS sum term, a constant 1 appears complemented. Hence this group gives $(A' + C')$.

Therefore: $F = (A + C)(A' + C')$

Note Memory rule: SOP -> group 1s. POS -> group 0s. In a POS group, constant 0 gives the normal variable and constant 1 gives the complemented variable.

5. Example with don't-care conditions

$F(A,B,C,D) = \text{Sum of minterms } (1,3,7,11,15) + d(5,9,13)$, where d denotes don't-care conditions.

AB \ CD	00	01	11	10
00	0 m0	1 m1	1 m3	0 m2
01	0 m4	X m5	1 m7	0 m6
11	0 m12	X m13	1 m15	0 m14
10	0 m8	X m9	1 m11	0 m10

A don't-care cell may be treated as 0 or 1, whichever helps obtain the simpler expression. Here we use m5, m9 and m13 as if they were 1s, allowing all eight cells in the middle two columns to be grouped together. Across this 8-cell group, A, B and C all change. Only $D=1$ remains unchanged.

Therefore: $F = D$

Note A don't-care does not automatically mean 1. It means the output for that input combination is irrelevant, so we may use the cell as 0 or 1 if it helps simplify the circuit.

Summary

No.	Problem	Simplified result	Main idea
1	Sum $m(3,4,6,7)$	$F = yz + xz'$	Extra legal group $m6, m7$ is redundant
2	Sum $m(1,2,3,5,6,7)$	$F = y + z$	Two overlapping groups of four
3	4-variable SOP	4 equally minimal forms	Different valid group choices
4	POS example	$F = (A+C)(A'+C')$	Group the 0s
5	Don't-care example	$F = D$	Use X cells only when helpful

For Problem 3, the four forms are:

$$F = B'D' + BD + AB' + B'C$$

$$F = B'D' + BD + AB' + CD$$

$$F = B'D' + BD + AD + B'C$$

$$F = B'D' + BD + AD + CD$$